



RKDF UNIVERSITY RANCHI

MASTER OF SCIENCE (M.Sc.) ENVIRONMENTAL SCIENCE

DISTRIBUTION OF DIFFERENT SUBJECTS AND CREDITS IN VARIOUS SEMESTERS

SEMESTER-I		
Subject Code	Subject Name	Credits
PEVS101	Foundation Course in Ecology	4
PEVS102	Earth and its Atmosphere	4
PEVS103	Aquatic Environment	4
PEVS104	Environmental Microbiology and Biotechnology	4
PEVS155	Practical- I Lab*	4
	Total	20
SEMESTER-II		
PEVS201	Biodiversity and Conservation	4
PEVS202	Energy Resources and Conservation	4
PEVS203	Environmental Pollution and Toxicology	4
PEVS204	Environmental Monitoring and Management	4
PEVS255	Practical- II lab*	4
	Total	20
SEMESTER-III		
PEVS301	Biostatistics and Modeling	4
PEVS302	Water Resource Management	4
PEVS303	Society and Environment	4
PEVS304	Remote Sensing and GIS	4
PEVS355	Practical-III Lab*	4
	Total	20
SEMESTER-IV		
PEVS401	Disaster Management	4
PEVS402	Environmental Legislation	4
PEVS433	Practical-IV lab*	4
PEVS444	Dissertation /Industrial Training/Project Work	8
	Total	20
	Grand Total	80

*Syllabus for the all practical papers will be based on the theory syllabus having practical implications.

SEMESTER-I

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Foundation Course in Ecology	PEVS101

1. Organisms and Environment: Holocoenotic nature of environment; abiotic and biotic environment.
2. Ecological adaptations: Morphological and physiological responses of organisms to temperature and water.
3. Population ecology: Population characteristics, population growth, carrying capacity, population regulation, life history strategies (r and K selection), population interactions including Lotka – Volterra model, population differentiation.
4. Community ecology: Concepts of community and continuum; community attributes; species diversity (α , β and γ); community coefficients; concept of ecological niche.
5. Community development: Models and mechanisms of ecological succession; changes in ecosystem properties during succession; Concept of climax.
6. Ecosystem organization: Ecosystem structure and functions, primary production (methods of measurement, global pattern, controlling factors); energy dynamics (trophic organization, energy flow pathways, ecological efficiencies); litter fall and decomposition; mineral cycles in terrestrial and aquatic ecosystems
7. Ecosystem management: Concepts; sustainable development; sustainability indicators.

Suggested Readings:

1. E.P.Odum and G.W.Barrett.2005.Fundamentals of Ecology.Cengage Learning India Pvt. Ltd.
2. J.S.Singh, S.P.Singh and S.R.Gupta.2008.Ecology, Environment & Resource Conservation. Anamaya Publications.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Earth and its Atmosphere	PEVS102

1. Weathering and erosion processes; Types and formation of soils and soil profile. Earthquakes, Volcanoes, Landslides and Floods; and their impact on environment.
2. Major rock and ore forming minerals: Properties of minerals; Igneous, sedimentary and metamorphic rocks. Impact of mining on environment.
3. Groundwater: Occurrence; Saltwater intrusion; Pollution and management.
4. Evolution of the earth's atmosphere, composition and thermal stratification, atmosphere and the earth's radiation balance, circulation of atmosphere, atmospheric stability, lapse rates and mixing heights, plume behavior; Gaussian plume model; Photochemistry of nitrogen oxides, oxygen, ozone and chlorides in the atmosphere.
5. Köppen's climate classification system; General relationship between landscape, biomes and climate.

Suggested Readings:

1. T.R.Oke.2006.Boundary layer climates.Methuen & Co.Ltd.
2. S.Pal Arya.2001.Introduction to Micrometeorology.Academic Press.
3. H.R.Byers.2006.General Meteorology.McGraw-Hill.
4. K.S.Valdiya.1987.Environmental Geology.Tata McGraw-Hill.
5. J.M.Wallace and P.V.Hobbs.2006.Atmospheric Science – An introductory survey.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Aquatic Environment	PEVS103

1. Diversity of aquatic habitats; hydrologic cycle
2. Aquatic food webs including microbial loop; trophic cascade
Measurement of aquatic primary productivity
3. Lakes-Origin and classification, ecological zonation, thermal stratification, water circulation, physical and chemical characteristics
4. Phytoplankton – diversity and models of nutrient-limited growth, paradox of plankton; a general account of zooplankton
5. A general account of benthic and periphytic communities Characteristics of running water habitats; river continuum concept
6. Oceans: Chemistry of seawater, circulation and ecological zonation in sea, marine biota, coral reefs
7. A general account of estuaries and wetlands Eutrophication: Causes, consequences and control measures

Suggested Readings:

1. Dobson, M. and Frid, C. 1998. Ecology of Aquatic Systems. Longman.
2. Adams, S.M. (Ed). 2002. Biological Indicators of Aquatic Ecosystem Stress. American Fisheries Society, Bethesda.
3. Talling, J.F. and Lemoalle, J. 1998. Ecological Dynamics of Tropical Inland Waters. Cambridge University Press.
4. Wetzel, R.G. and Likens, G.E. 2000. Limnological Analysis. Springer-Verlag.
5. Wetzel, R.G. 2000. Limnology: Lake and River Ecosystems. Academic Press.
6. Dodson, S. 2005. Introduction to Limnology. McGraw-Hill, New York.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Environmental Microbiology and Biotechnology	PEVS104

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1. Introduction to microorganisms: General characteristics, nutritional types, microbial diversity.
2. A brief idea of techniques relating to isolation, purification and culture of microorganisms.
3. Types of interaction between plants and microbes.
Microorganisms and soil fertility.
4. Microorganisms in extreme environments.
Microbial toxins and environmental hazards.
5. Brief account of plant diseases and their ecosystem level effects.
6. Microbes and public health: Brief account of microbial diseases in humans. Microbially induced corruptions and biofilms.
7. Bioremediation of organic and inorganic contaminants
8. Strategies of recombinant DNA technology and its applications.
9. Release of genetically engineered microorganisms: safety and environmental risks.
10. Vermicular and bio-fertilizer technology.

Suggested Readings:

1. Raina M. Maier. 2000. Environmental Microbiology. Academic Press.
2. Pepper, I. and C. P. Gerba. 2004. Environmental Microbiology (2nd Edition). Academic Press.

SEMESTER-II

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Biodiversity and Conservation	PEVS201

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1. Introduction to biodiversity: species, genetic and ecosystem diversity.
2. Biodiversity magnitude and distribution: diversity gradients and related hypotheses, biodiversity and ecosystem function, methods for biodiversity monitoring.
3. Biodiversity and ecosystem services: provisioning, regulating, cultural and supporting.
4. Threats to biodiversity: Natural and anthropogenic, species extinctions, IUCN threat categories, Red data book, Invasions: causes and impact.
5. Biodiversity conservation, principles and strategies; in-situ and ex-situ conservation, Protected Area Network.
6. Biodiversity Hotspots: concepts, distribution and importance.
7. Use of biodiversity: Source of food, medicine, raw material, aesthetic and cultural. Biodiversity prospecting.

Suggested Readings:

1. Anne E. Magurran. 2003. Ecological diversity and its measurements. Blackwell Publications.
2. J.S.Singh, S.P. Singh and S.R. Gupta. 2008. Ecology, Environment and Resource Conservation. Anamaya Publications (New Delhi).
3. V.H. Heywood and Watson R.T. (Ed). 1995. Global Biodiversity Assessment: UNEP. Cambridge University Press.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Energy Resources and Conservation	PEVS202

1. Introduction: Energy, work and power; Energy and people; Energy sources – Resource and reserves - an overview; an overview of the current global and National Energy Scenario.
2. Fossil Fuels: Oil, coal, natural gas, shale, tar sands – Sources, exploration, exploitation; environmental consequences.
3. Nuclear Energy: Nuclear fission and Fusion; Nuclear fuel cycle, Nuclear reactors (PWR, BWR, Gas Cooled Breeder) and nuclear power.
4. Renewable and Alternative Energy Sources: Solar energy, solar power, photovoltaic cells; Wind power; Geothermal energy; Ocean energy; Fuel cells.
5. Bio Energy: Biomass conversion processes; Biodiesel; Environmental consequences of biomass resource harnessing.
6. Energy Conservation: National energy policy, energy efficiency improvement, audit and energy saving.

Suggested Readings:

1. M. Dayal. (6th Ed). 1997. Renewable Energy: Environment and Development. Konark Pub. Pvt. Ltd.
2. S.Vandana.2002.Alternative Energy.APH Publishing Corporation.

3. S. K. Agarwal. 2003. Nuclear Energy: Principles Practice and Prospects. APH Publishing Corporation.
4. P. Chaturvedi. 1995. Bio-Energy Resources. Concept Publications.
5. V.S. Mahajan. 1991. National Energy: policy, crisis and growth. Ashish Publishing House.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Environmental Pollution and Toxicology	PEVS203

1. Air pollution: Types and sources, Effects of SO₂, NO₂, O₃, HF, photochemical smog and particulates on plants and human health, aeroallergens and allergies.
2. Ozone layer depletion: Causes and consequences.
3. Noise pollution: Types, sources and effects on human health.
4. Water Pollution: Types and sources; Effects on water quality, plants and human health; Thermal pollution.
5. Soil pollution: Types and sources, Effects of pesticides and heavy metals on ecosystems, mechanisms of metal toxicity, metallophytes.
6. Radioactive pollution: Sources and hazards. Solid waste: Sources and effects.
7. Toxicology: Principles of toxicology, dose-response relationships, Chronic and acute toxicity; Effective concentration, LD₅₀, Median tolerance limit and Margin of safety; Toxicity testing (Holistic and Numeric approach).
8. Uptake, bioaccumulation, bio-transformation and excretion of xenobiotics. Role of temperature and humidity in human health.

Suggested Readings:

1. A.K. De. (3rd Ed). 2008. Environmental Chemistry. New Age Publications India Ltd.
2. I. C. Shaw and J. Chadwick. 1997. Principles of Environmental Toxicology. Taylor & Francis Ltd.
3. S.C. Santra. 2011. Environmental Science. New Central Book Agency.
4. Ira. S. Richards. 2008. Principles and Practices of Toxicology in Public Health. Jones and Barlett Publications.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Environmental Monitoring and Management	PEVS204

1. Ambient air monitoring; Methods of collection and analyses of gaseous and particulate pollutants.
2. Methods of collection of water samples and analyses of physico-chemical characteristics. Bio-monitoring and bio-indication.
3. Principles of chromatography, spectrophotometry, electro-analytical and radio-analytical techniques.
4. Environmental Management: Principles and strategies ; Indicators of environmental quality, economic valuation; pipeline model; closed loop model and material balance model; environmental cost-benefit analysis; sources of uncertainty in cost and benefit estimates; Fiscal incentives in pollution control and management.
5. Environmental management system (EMS): ISO-14000; Environmental audit; Environmental clearance for establishing industries; Environmental Impact Assessment (EIA); EIA guidelines 1994, Environmental taxes.
6. International trade and environment; Trade Related Intellectual Properties (TRIPs),

Intellectual Property Rights (IPRs), Corporate environmental ethics.

Suggested Readings:

1. C. N. Sawyer, P. L. McCarty and G. F. Parkin. 2002. Chemistry for Environmental Engineering and Science. John Henry Press.
2. H. H. Rump. 2000. Laboratory Manual for the Examination of Water, Waste water and soil. Wiley-VCH.
3. R. K. Saprú. 1987. Environmental Management in India (Vol. I & II). Ashish Publishing House.

SEMESTER-III

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Biostatistics and Modeling	PEVS301

1. General concepts and terminology; sampling methods; distribution of attributes; tests of hypothesis and significance; contingency tables and chi-square test; comparison of means: t-test, multiple range tests; Simple experimental design and analysis of variance; correlation and regression analysis; Introduction to multivariate methods.
2. Approaches to development of models; conceptual, statistical and mathematical models; steps in the modeling process; single and multiple regression models; modeling coupled human-natural systems; model testing and validation; models as predictive tools.

Suggested Readings:

1. S.C.Gupta and V.K.1986.Kapoor.Fundamentals of Mathematical Statistics.S. Chand & Co.
 2. Aslam Mahmood.1993.Statistical Methods in Geographical Studies.Rajesh Publications, New Delhi.
 3. J.Medhi. Statistical Methods: 1992. An Introductory Text. New Age International Ltd. Publishers.
- Gupta, S.P.2000. Statistical Methods. Sultan Chand & Sons, New Delhi

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Water Resource Management	PEVS302

1. Global distribution of water resources, water need and consumption; Threats to surface water resources; Principles and approaches to surface water management. Watershed management: Rain water harvesting and storage, recharging of ground water; role of dams.
2. Properties of sewage and industrial effluents; effluent standards; treatment of industrial effluents, sewage treatment (primary, secondary and tertiary treatment), advanced treatments (nitrate and phosphate removal); Sludge treatment and disposal; Waste water use.
3. Drinking water quality and water treatment (desalination, ion-exchange, reverse osmosis and disinfection of water).

Suggested Readings:

1. A.J.Schleiss and R.M.Boes.2011.Dams and Reservoirs under Changing Challenges. CRC Press.
2. J.N.Parkinson, J.A. Goldenfum and C.E.M.Tucci.2010.Integrated Urban Water Management. CRC Press.
3. A.N.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Society and Environment	PEVS303

1. Social perspectives of environment: Global and Indian issues. Sustainable development: Concept, components and strategies.
2. Social impacts of growing human population and affluence, food security, hunger, poverty, malnutrition, famine.
3. Social impacts of water crisis, global climate change, O₃ depletion, nuclear accidents, acid rain, consumerism and waste products.
4. Problems related to major dams and other developmental projects, resettlement and rehabilitation.
5. Environment and human health: epidemiological issues, women and child welfare, family welfare programme.
6. Environmental education, Environmental ethics, public awareness, peoples participation in resource conservation and environmental protection.

Suggested Readings:

1. S.C.Naik and T.N.Tiwari. 2006. Society and Environment. Oxford & IBH Publishers.
2. S.C.Santra. 2011. Environmental Science. New Central Book Agency.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Remote Sensing and GIS	PEVS304

1. Electromagnetic Radiation as Remote Sensing Medium; General Mechanism of Remote Sensing Data Recording; General Characteristics of Remote Sensing Platforms; General Characteristics of Remote Sensing Sensors; Indian Remote Sensing Satellites and Sensors.
2. Spectral Characteristics of Common Natural Objects; Atmospheric Effects on Remote Sensing Data; Spectral Signatures and Spectral Response Patterns; Resolution of Remote Sensing Data; Characteristics of Raw Remote Sensing Data
3. Nature of Qualitative Information and Sequence in Interpretation; Elements of Image Patterns-Landforms, Drainage, Erosion Details; Applications of Remote Sensing; Remote Sensing Applications in Environmental Studies; Digital Image enhancement and classification methods; Principles of Microwave Remote Sensing; Characteristics of Microwave remote sensing Data; Radar and Lidar: Applications of Microwave Remote Sensing Data.
4. Geographical Data and GIS; Coordinate Systems and Datums; Digital representation of geographical data-Raster and Vector models; GIS Data Standards-Concepts and Components; Conceptual and Logical Data Modeling; Applications of GIS

Suggested Readings:

1. Burrough, P.A. and McDonnell, R. 1998. Principles of Geographical Information Systems. Oxford University Press, NY.
2. Campbell, J.B. (2nd Ed), 1996. Introduction to Remote Sensing. Taylor and Francis.
3. Christopher, J. 1997. Geographical Information Systems and Computer Cartography. Longman.

SEMESTER-IV

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Disaster Management	PEVS401

1. Understanding hazards and disasters, hazards to environment, risk Assessment, vulnerability analysis.
2. Dimensions of disaster, disaster impact assessment; types of disaster impact; disaster trends and patterns; flood, drought, cyclone, tsunami, earthquake and volcanoes & its management.
3. Understanding natural and man-made disaster; disaster preparedness; disaster responses; reducing the impacts of disaster.
4. Biophysical hazards; disaster due to diseases; disaster in medicines; disaster aid; Community preparedness; forecasting and warning; rehabilitation; reconstruction and recovery; slope Instability and landslide hazard.

Suggested Readings:

1. G. F. White (Ed). 1974. Natural Hazards – Local, National, Global. Oxford University Press.
2. V.T.Chow.1964.HandbookofAppliedHydrology.McGraw-Hill.
3. A.N.StrahlerandA.H.Strahler.1973.EnvironmentalGeoscience-InteractionBetween Natural Systems and Man . Santa Barbara, California: Hamilton Publishing.
4. P. Reining. 1978. Handbook of Desertification Indicators. Washington D.C.: American Association for the Advancement of Science.
5. K.S.Valdiya.1987.EnvironmentalGeology.TataMcGraw-Hill.

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Environmental Legislation	PEVS402

1. Fundamental right to clean environment and duties; State's obligation. Power of Parliament to legislate environmental legislations.
2. Definition: Environment, air pollution, water pollution, hazardous substances, handling, animal article, forest, non-forest purposes, biodiversity.
3. Environmental legislation enforcement authorities: (i) Under the Water and Air Act-composition, powers and functions, and (ii) Under the Environment (Protection) Act, 1986- powers Environmental dispute redress bodies:(i)National Green Tribunal composition and jurisdiction,
4. (ii) Trial court- jurisdiction under the environmental legislations, and (iii) Executive Magistrate's powers
5. The Forest (Conservation) Act 1980: Objectives and Mechanism.
6. The Wild Life (Protection) Act 1972: Objectives, prohibition on hunting and trade. The Biodiversity Act 2002: object and National Biodiversity Authority.
7. Convention on Climate Change: objectives, principles and commitment-India's response.

Suggested Readings:

1. Environmental Laws.2005.Universal Law Publishing.
2. S.DiwanandA.Rosencranz.2005.Environmental Law and Policy in India. Oxford Univ. Press

Course	Subject Title	Subject Code
M.Sc. Environmental Science	Dissertation /Industrial Training/Project Work Report	PEVS444

Students shall be required to go for Industrial training (8 credits) of four weeks during summer vacation after completion of Semester II examination. The training will be evaluated during Semester IV on the basis of a report and presentation.

